

## AI Capex and the Corporate Credit Market: Evidence on Crowding Out, 2010–2026

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**Abstract.** We ask whether the rapid increase in AI, datacenter, and hyperscaler financing since 2023 is changing the allocation, price, and availability of credit for other corporate borrowers. Measured AI-side investment, defined as cash capital expenditure of the four hyperscalers plus Oracle and CoreWeave, was \$389 billion in 2025 (Observed, A), up from \$24 billion in 2015, with about \$789 billion guided for 2026. Against that boom, US corporate credit aggregates expanded rather than contracted: bond issuance averaged \$1.90 trillion per year in 2023–2025 versus \$1.68 trillion in 2015–2022 (Observed, A), bank credit reached \$19.8 trillion in August 2026, and credit spreads narrowed. In the euro area, total bank lending to non-financial corporations grew 1.7 percent per year in 2023–2026 versus 2.0 percent in 2015–2022, and the reconstructed credit premium was flat at about 1.9 percentage points (Reconstructed, B). What the data do show is sectoral divergence rather than aggregate rationing: euro-area manufacturing loan stocks contracted after 2022 while real estate and professional services kept growing, technology borrowers absorbed a rising share of growing US credit, and AI financing increasingly flows through private credit, project vehicles, and foreign-currency bond markets that sit partly outside measured bank aggregates. Every result is descriptive; the samples are short and the post-2023 window coincides with the monetary tightening cycle, so nothing here identifies a causal effect of AI investment on credit conditions.

**Keywords:** AI infrastructure, corporate credit, crowding out, capital allocation, private credit

**JEL Classification:** E51, G21, G32

# Contents

|          |                                                                      |           |
|----------|----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                  | <b>3</b>  |
| <b>2</b> | <b>Institutional background</b>                                      | <b>4</b>  |
| <b>3</b> | <b>Related literature</b>                                            | <b>4</b>  |
| <b>4</b> | <b>Data</b>                                                          | <b>6</b>  |
| <b>5</b> | <b>Methodology</b>                                                   | <b>6</b>  |
| <b>6</b> | <b>Results</b>                                                       | <b>8</b>  |
| 6.1      | The AI-side investment and financing boom . . . . .                  | 8         |
| 6.2      | US credit aggregates expanded alongside the boom . . . . .           | 9         |
| 6.3      | Euro-area: quantities slowed modestly, prices did not rise . . . . . | 10        |
| 6.4      | Spreads narrowed while AI capex accelerated . . . . .                | 11        |
| 6.5      | Where the marginal credit went . . . . .                             | 11        |
| 6.6      | The private-credit and off-balance-sheet channel . . . . .           | 12        |
| 6.7      | Cross-currency transmission . . . . .                                | 12        |
| <b>7</b> | <b>Limitations and robustness</b>                                    | <b>12</b> |
| <b>8</b> | <b>Conclusion</b>                                                    | <b>13</b> |
| <b>A</b> | <b>Data appendix</b>                                                 | <b>16</b> |

# 1 Introduction

The question is simple and the answer matters for how capital gets priced. Since 2023, a small number of technology companies have committed to the largest corporate investment program in modern history: datacenters, power procurement, and AI compute. The four hyperscalers (Amazon, Microsoft, Alphabet, Meta) plus Oracle and CoreWeave spent \$389 billion in cash capital expenditure in 2025 (Observed, A), up from \$24 billion in 2015, and company guidance puts 2026 near \$789 billion. The financing has scaled with it: those companies issued about \$108 billion in public bonds in 2025 and about \$214 billion in the first nine months of 2026 (Observed, A), then began raising equity at record scale.

The concern is crowding out. Corporate credit is not an unlimited pool. If a handful of borrowers absorb hundreds of billions in bank loans, bond-market capacity, and private-credit commitments, other borrowers may face slower credit growth, wider spreads, shorter maturities, or tighter standards. The sectors most exposed are the capital-intensive ones that historically absorbed the marginal corporate credit: industrials and manufacturing, commercial real estate, utilities and power, energy infrastructure, telecommunications, transportation, healthcare, and construction.

This paper builds the sector-level evidence. We assemble a historical baseline covering 2010–2022 and compare it with the post-2023 AI investment cycle across the United States and the euro area, with the United Kingdom, Canada, and Asian markets where data permit. We measure bank-loan growth, bond issuance, lending standards, borrowing rates, credit spreads, syndicated lending, and private-credit deployment at the sector level, and we test whether anything changed after 2023 beyond what interest rates and the cycle explain.

The bottom line, stated first: the data do not show aggregate crowding out. US corporate credit expanded on every major measure while AI capex boomed, and spreads narrowed rather than widened. In the euro area, loan growth slowed modestly but the price of credit did not rise. What the data do show is divergence across sectors and across funding channels: manufacturing loan stocks contracted in the euro area while real estate kept growing; technology borrowers took a rising share of growing US syndicated credit; and much of the AI financing runs through private credit, off-balance-sheet project vehicles, and foreign-currency bond issuance that standard credit aggregates capture only partially. The financing boom is therefore better described as a reallocation across channels than as a rationing of the aggregate pool, though the opacity of the off-balance-sheet channel is itself a finding: roughly \$1.65 trillion of off-balance-sheet AI obligations at the five largest names (press-reported, Estimated, C) now sits outside the credit statistics most analysts watch.

Three caveats belong in the same paragraph as the headline. First, the post-2023 window coincides with the fastest monetary tightening cycle in four decades, and our samples are short: eleven annual US observations and an eight-sector euro panel. Nothing here identifies a causal effect of AI investment on credit conditions, and we do not claim one. Second, no public dataset isolates AI or datacenter borrowing inside sector aggregates; the capex series is total-company spending, an upper bound on AI investment. Third, 2026 observations are partial-year or company guidance and are marked as non-comparable throughout.

The rest of the paper proceeds as follows. Section 2 describes how corporate credit markets and AI financing channels actually work. Section 3 reviews the related literature. Section 4 documents the data, with a summary-statistics table. Section 5 sets out the methodology. Section 6 presents the results. Section 7 discusses limitations and robustness. Section 8 concludes. A data appendix defines every series.

## 2 Institutional background

Corporate borrowers fund themselves through four overlapping channels. Banks extend loans, including revolving facilities and term loans, and report them in aggregates such as the Federal Reserve H.8 and the ECB’s MFI balance-sheet statistics. Bond markets absorb public issuance, split between investment grade and high yield, tracked in the United States by SIFMA and in the euro area by the ECB’s securities-issues statistics. Syndicated loans, large facilities originated by groups of banks, are monitored through the Shared National Credit program in the United States. And private credit, non-bank direct lending by funds, has grown into a \$2.5 trillion global market as of 2025 (BIS, Observed, A), operating largely outside bank supervisory data.

AI infrastructure financing now uses all four channels, plus three more that complicate measurement. First, equity: Alphabet’s \$84.75 billion June 2026 raise, the largest technology equity raise on record (Observed, A), and Oracle’s \$45–50 billion 2026 funding plan, about half equity, show the funding mix shifting toward permanent capital that does not compete for credit-market capacity. Second, off-balance-sheet project vehicles: Meta’s Hyperion datacenter campus in Louisiana sits in a joint venture 80 percent owned by Blue Owl Capital funds, with about \$27 billion of senior secured notes issued by the project company, not by Meta (Observed, B). Meta discloses a residual value guarantee in footnotes, not as a liability. By September 2026 those notes traded at 94.4 cents on the dollar, the first secondary-market repricing of AI project risk. Third, cross-currency issuance: since April 2025 the hyperscalers have issued heavily in euros, sterling, and Canadian dollars. Alphabet has done €22.25 billion across three euro deals plus £5.5 billion including a 100-year tranche due 2126; Amazon has done €14.5 billion, £4.25 billion, and C\$14 billion (all Observed, A, from final term sheets). This transmits AI capital demand directly into European and other non-dollar funding pools.

The ECB’s August 2026 assessment of this last channel is the closest thing to an official verdict available: roughly €40 billion of hyperscaler euro bonds outstanding, just under 10 percent of gross new euro non-financial corporate issuance, with no clear crowding-out spillover to euro-area issuers so far, though some issuers reportedly avoided pricing on hyperscaler issuance days (ECB, Observed, B).

The sectors historically absorbing marginal corporate credit are the natural comparison group. In US syndicated lending, financials excluding banks and technology/telecom/media are the two largest non-bank borrower groups; in euro-area bank lending, real estate plus professional and administrative services is 41.7 percent of loan stocks, followed by manufacturing at 12.4 percent and wholesale and retail trade at 11.8 percent (ECB, Observed, A). If AI financing were rationing credit, these are the borrowers whose quantities, prices, or standards should show it first.

## 3 Related literature

The literature on credit crowding out supplies mechanisms and identification strategies directly relevant to the AI infrastructure cycle, but little of it studies that cycle itself. We organize it around three questions: how credit supply shocks propagate, how to measure them, and what the 2026 evidence says about AI financing directly.

The bank lending channel literature establishes that credit supply shocks travel through intermediary balance sheets and that borrowers differ in their ability to substitute across lenders. Bernanke and

Blinder (1988) gives the foundational statement of the quantity mechanism; Kashyap and Stein (2000) and Khwaja and Mian (2008) show the channel operates through bank balance sheets and that large firms offset lost bank credit through other markets while small firms cannot. Jiménez et al. (2012) identify the bank balance-sheet channel with loan-application data, and Amiti and Weinstein (2018) attribute 30 to 40 percent of aggregate loan and investment fluctuations in Japan to idiosyncratic bank supply shocks. Chodorow-Reich (2014) finds that credit withdrawal explains between one third and one half of the employment decline at bank-dependent firms after the Lehman bankruptcy. The implication for our question is that if AI financing binds intermediary balance sheets, the cost falls disproportionately on borrowers that cannot substitute into bond or private-credit markets.

A second strand documents misallocation when scarce credit is steered toward the wrong borrowers. Caballero et al. (2008) and Peek and Rosengren (2005) show that forbearance toward weak borrowers congests markets and depresses restructuring; McGowan et al. (2017) and Banerjee and Hofmann (2018) extend the zombie-firm result across advanced economies. In the European sovereign debt crisis, domestic banks absorbed sovereign issuance at the expense of corporate lending (Becker and Ivashina, 2018; Ongena et al., 2019; Marco, 2019), with spillovers toward sovereigns and low-quality relationship borrowers (Crosignani et al., 2017; Broner et al., 2013; Acharya et al., 2019). Chakraborty et al. (2018) provide the closest analogue to a sectoral AI shock: banks more exposed to housing booms expanded mortgage lending while cutting commercial lending, and affected firms invested less. Gopinath et al. (2017) add the caution that abundant cheap credit can worsen allocation, and Kalemli-Özcan et al. (2022) quantify how debt structure and rollover risk propagate credit shocks to investment. Greenwood et al. (2010) show that corporate issuance responds to the funding gaps left by large borrowers, a mechanism directly relevant when hyperscaler supply hits bond markets, while Adrian et al. (2013) show that bond markets can absorb part of a bank credit contraction, so total financing need not fall when bank credit does.

The survey literature provides the instruments to detect crowding out in real time. Lending officer surveys separate loan supply from demand (Lown and Morgan, 2006; Bassett et al., 2014; Maddaloni and Peydró, 2011; Ciccarelli et al., 2015), and we use the Federal Reserve SLOOS and the ECB Bank Lending Survey on that basis. Dell’Ariccia et al. (2012) document how credit standards deteriorate over the cycle, the baseline for spotting overheated lending to any sector, and López-Salido et al. (2017) give a cyclical timing device for judging whether current credit conditions reflect overheating. For the cross-currency dimension, Liao (2020) show that currency-hedged borrowing-cost differentials explain up to one third of cross-currency issuance flows, and the ECB’s 2025 review of reverse-yankee bonds finds unhedged cost differences explaining up to 40 percent of issuance-share variation (European Central Bank, 2025): AI issuers can migrate across currencies, broadening the pool rather than draining one market.

The closest direct evidence on AI financing comes from 2026 BIS and ECB publications. Aldasoro et al. (2026) document the financing pivot: as capex outgrew internal cash flows, hyperscalers turned to bond markets and private credit, with construction, power, and tenant-concentration risks pushing some financing into bespoke structures. A BIS Quarterly Review box (Bank for International Settlements, 2026) reports hyperscaler gross bond issuance topping \$100 billion in 2025, CDS spreads rising on supply, and off-balance-sheet joint ventures creating debt-like “shadow borrowing” off corporate balance sheets. Duquerroy et al. (2026) find US technology giants increasingly tapping the euro-area bond market, raising the risk that such inflows crowd out other issuers. Meta Platforms, Inc. (2025) is the primary record of the Hyperion joint venture, the template for off-balance-sheet AI project finance.

On the private-credit channel, International Monetary Fund (2024b) assess the opacity and allocation risks of the channel now funding part of the AI buildout; Avalos et al. (2025) size global direct lending above \$1.2 trillion with more than 87 percent originated in the United States; and Abbassi et al. (2026) argue private credit is structurally suited to asset-heavy, bespoke AI projects that sit awkwardly in traditional bank and bond channels. McKinsey & Company (2025) estimate about \$5.2 trillion of AI-related datacenter investment through 2030 in their central scenario, sizing the investment impulse whose financing the crowding-out question concerns; Goldman Sachs (2024) frame the demand-side uncertainty around those returns; International Monetary Fund (2024a) survey AI’s implications for capital-market activity.

The gap this paper fills is sector-level, pre-versus-post-2023 evidence. The mechanisms literature is well identified but studies different episodes: sovereign crises, housing booms, monetary shocks. AI issuers are investment grade, often cash rich, and borrow across bond markets, private credit, project entities, and foreign currencies, spreading pressure across funding pools rather than binding one bank balance sheet. No prior study compares sector-level credit quantities, prices, and standards before and after the AI capex cycle across the United States and euro area with measured AI-side financing. That comparison is this paper’s contribution.

## 4 Data

We use primary datasets wherever possible. Table 1 summarizes the core series; the data appendix defines every series used.

US data come from the Federal Reserve (H.8 bank credit, SLOOS lending standards, H.15 interest rates, Z.1 flow of funds), SIFMA corporate bond issuance statistics, and the Shared National Credit program. Euro-area data come from the ECB Statistical Data Warehouse (MFI balance-sheet statistics, bank lending survey, MFI interest-rate statistics, securities-issues statistics) and the BIS (total credit, international debt securities). UK data come from the Bank of England; Canadian and Asian coverage from the Bank of Canada, the World Bank WDI, and BIS publications, supplemented by archived press and industry sources for datacenter project finance. AI-side investment is built from hyperscaler 10-K and 10-Q filings, SEC final term sheets for bond deals, and company announcements for equity raises and joint ventures. Every raw pull is archived in the project workspace; processing scripts are re-runnable.

Coverage runs 2010–2026 where sources permit, with estimation windows of 2015–2022 (baseline) versus 2023–2026 (AI cycle) for the core comparisons. Data-through dates differ by source and are recorded in the data appendix. Known breaks: the SNC \$100 million reporting threshold from 2018, euro-area composition changes, and the ECB securities-issues mirror ending in March 2022, which removes any 2023–2026 euro-area bond-financing comparison from that source.

## 5 Methodology

The empirical strategy is comparative and descriptive. For each credit metric we compute the baseline-window mean (2015–2022) and the AI-cycle-window mean (2023 through latest available) and test whether the difference survives basic controls. We do not estimate a structural model of credit allocation, and we state the identification limit plainly in Section 7: the post-2023 indicator

**Table 1.** Summary statistics: core credit and AI-investment series

| Series                          | Unit      | Period     | Mean  | Latest                    |
|---------------------------------|-----------|------------|-------|---------------------------|
| <i>AI-side investment</i>       |           |            |       |                           |
| Hyperscaler cash capex          | USD bn/yr | 2015–2025  | 142.9 | 389.0 (2025)              |
| Hyperscaler bond issuance       | USD bn/yr | 2025–2026  | —     | 108 (2025); 214 (9M 2026) |
| <i>United States</i>            |           |            |       |                           |
| Bank credit (H.8)               | USD tn    | 2015–2026  | 15.6  | 19.8 (Aug 2026)           |
| C&I loan growth                 | % yoy     | 2015–2026  | 3.7   | 8.6 (YTD Aug 2026)        |
| Corporate bond issuance (SIFMA) | USD bn/yr | 2015–2025  | 1,743 | 2,219 (2025)              |
| NFC net borrowing (Z.1)         | USD bn/yr | 2015–2025  | 606   | 711 (2025)                |
| SLOOS net tightening            | net %     | 2015–2026  | 5.0   | 4.5 (2026-Q3)             |
| US IG spread                    | pp        | 2021–2025  | 1.2   | 1.04 (2023–25 mean)       |
| US HY spread                    | pp        | 2021–2025  | 4.0   | 3.56 (2023–25 mean)       |
| <i>Euro area</i>                |           |            |       |                           |
| MFI loans to NFCs               | EUR tn    | 2015–2026  | 4.9   | 5.5 (Aug 2026)            |
| NFC loan growth                 | % yoy     | 2015–2026  | 1.9   | 1.7 (2023–26 mean)        |
| BLS net tightening              | net %     | 2015–2026  | 3.2   | 8.5 (2023–26 mean)        |
| NFC lending rate                | %         | 2015–2026  | 2.6   | 4.32 (2023–26 mean)       |
| Reconstructed credit premium    | pp        | 2015–2026  | 1.9   | 1.86 (2023–26 mean)       |
| <i>United Kingdom</i>           |           |            |       |                           |
| M4 net lending                  | GBP bn/yr | 2015–2025  | 72    | 142.9 (2025)              |
| PNFC bond net issuance          | GBP bn/yr | 2015–2025  | -3.4  | -26.7 (2025)              |
| <i>Private credit</i>           |           |            |       |                           |
| Global private credit stock     | USD tn    | 2023, 2025 | —     | 2.5 (2025, BIS)           |
| Tech share of US direct lending | %         | 2010/19/25 | —     | 44 (2025, BIS)            |

**Notes:** Means are computed over the stated period from archived series. AI capex is the four-hyperscaler plus Oracle and CoreWeave cash total (Observed, A). Spreads are reconstructed as yield minus risk-free rate (Reconstructed, B). 2026 values are partial-year or guidance where marked in the text. **Source:** Federal Reserve, ECB, Bank of England, SIFMA, BIS, company filings; see data appendix.

coincides with the monetary tightening cycle, the energy-price shock unwind, and the AI capex boom, and no specification below separates them.

Three computations carry the analysis. First, a euro-area sector panel: annual sector loan growth (eight NACE sectors, 2015–2025, 88 observations) regressed on a post-2023 indicator, the ECB AAA 5-year par yield as the rate control, and sector fixed effects, with standard errors clustered by sector. Interactions for manufacturing and construction test sectoral divergence from the pooled reference group. Second, US annual time series (2015–2025, 11 observations): SIFMA corporate bond issuance, H.8 C&I loan growth, and Z.1 NFC net borrowing each regressed on the 10-year Treasury yield, with a Chow test for a break at 2023 and a residual check that fits the model on 2015–2022 and compares 2023–2025 actuals to predictions. Third, spread decompositions: US investment-grade and high-yield spreads reconstructed as SIFMA yield-to-maturity minus the 10-year Treasury, and a euro-area credit premium reconstructed as the NFC new-business composite lending rate minus the ECB AAA 5-year par yield, each compared across windows and correlated with AI capex growth.

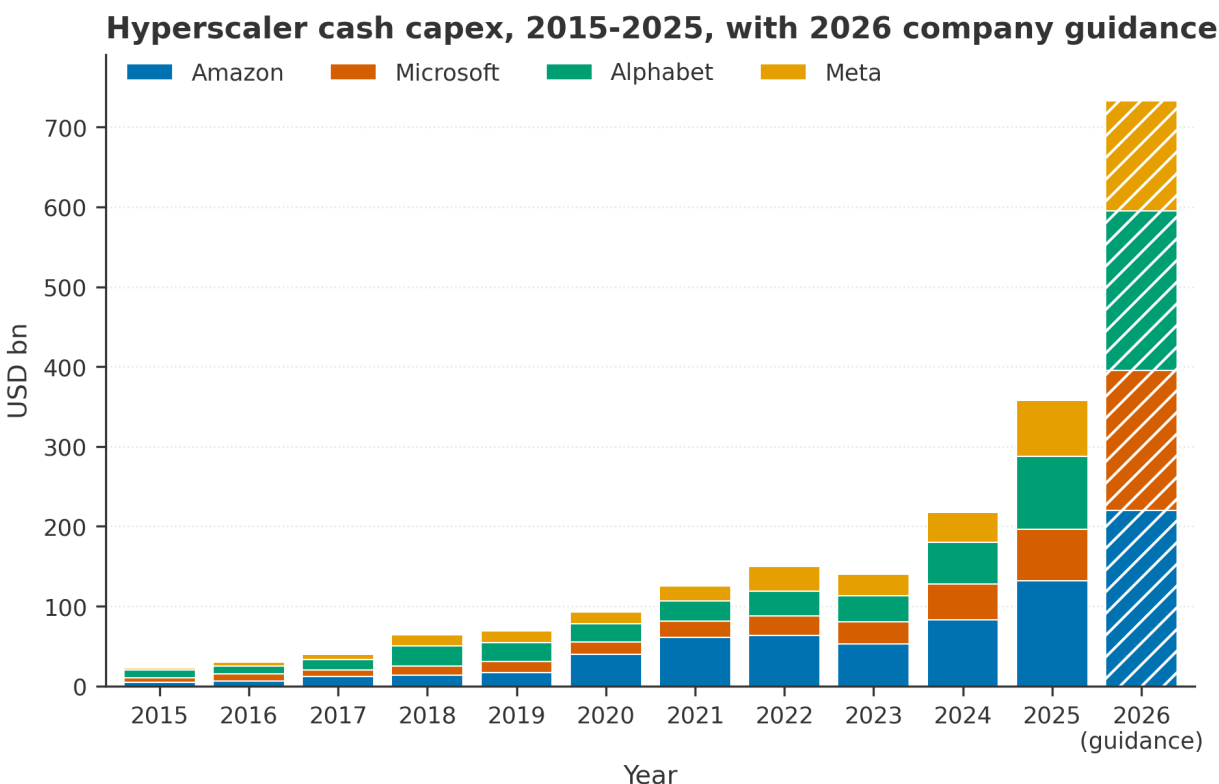
A marginal-absorption arithmetic completes the picture: for each region we compute which sectors absorbed the change in credit stocks or issuance between the two windows, so the reader can see

where the marginal dollar went. All processing code is archived and re-runnable; every number below traces to an archived series.

## 6 Results

### 6.1 The AI-side investment and financing boom

Figure 1 shows the scale. Hyperscaler cash capex rose from \$24 billion in 2015 to \$389 billion in 2025 (Observed, A), compounding at 31.9 percent per year, with company guidance putting 2026 near \$789 billion. The 2023 dip, when Amazon and Meta both cut, is the only down year in the series; the two-year increase from 2023 to 2025 (\$217 billion) exceeded the combined capex of 2015 through 2018. Two warnings bound the series: it is total-company cash capex, not AI-attributable spending, so it is an upper bound on AI investment; and 2026 is guidance, not actuals, with Microsoft’s figure including finance leases on a non-comparable definition.

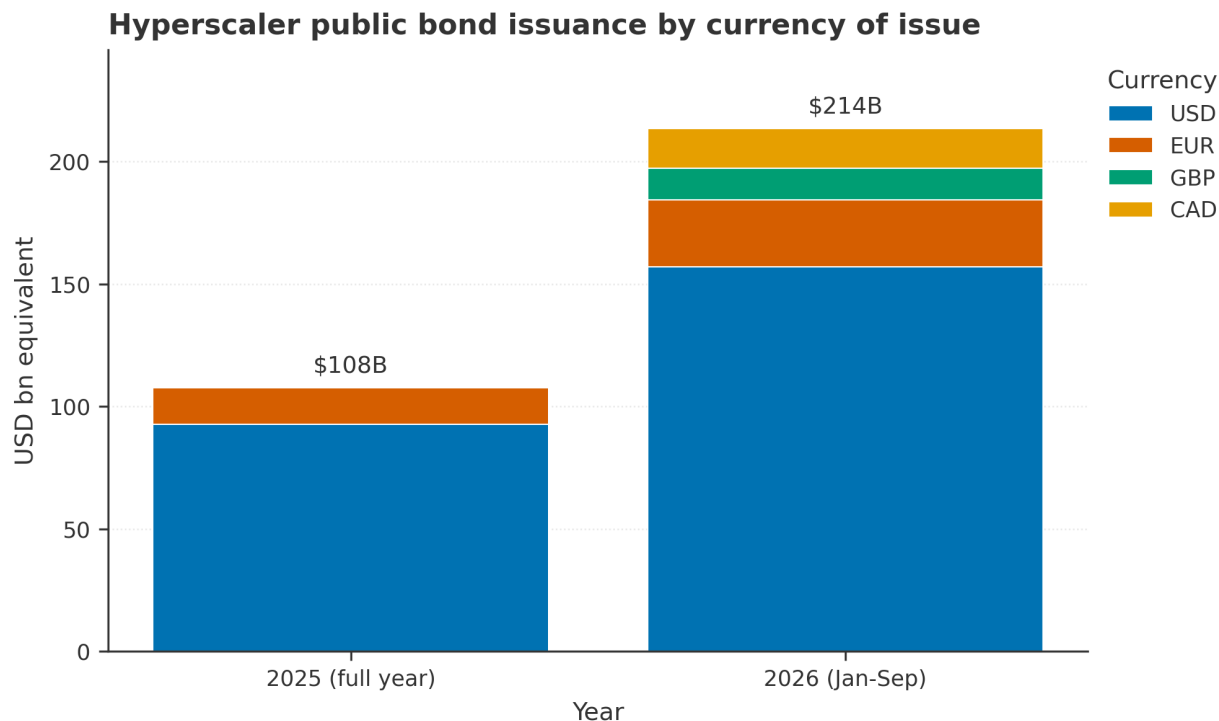


**Figure 1.** Hyperscaler cash capex, 2015–2025, with 2026 company guidance

**Note:** Bars show reported cash capex for 2015–2025. The 2026 bar (hatched) shows company-issued guidance, not actuals, and is not comparable to the cash-capex history: Microsoft’s 2026 guidance includes finance leases. Data through 2025 actuals; guidance as issued in 2026. **Source:** Company filings via `data/ai-capex/ai_infrastructure_investment_annual.csv`. Observed, A.

The financing, shown in Figure 2, scaled with the investment and then diversified. The five names issued about \$108 billion in public bonds in 2025 and about \$214 billion in the first nine months of 2026 (Observed, A, from SEC final term sheets). The cross-currency dimension is new: before 2025 none of the five had material euro issuance in our sample, while since April 2025 Alphabet has

done €22.25 billion and Amazon €14.5 billion, both adding sterling and Canadian-dollar tranches. In 2026 the mix shifted toward equity: Alphabet’s \$84.75 billion June raise and Oracle’s \$45–50 billion plan (about half equity) fund AI capex with permanent capital that does not compete for credit-market capacity. That mix shift is itself a datum for the crowding-out question.



**Figure 2.** Hyperscaler public bond issuance by currency of issue

**Note:** 2025 is a full year; 2026 covers January–September only and is not comparable to the full-year 2025 bar. Totals: \$108 billion (2025), \$214 billion (2026 Jan–Sep). Converted to USD equivalents at issue-date rates. **Source:** data/ai-capex/hyperscaler\_public\_bond\_issuance\_by\_currency.csv. Observed, A.

## 6.2 US credit aggregates expanded alongside the boom

Figure 3 shows the three US aggregates. Bank credit reached \$19.8 trillion in August 2026 (Observed, A); annual growth averaged 6.6 percent in 2015–2022, dipped to -0.9 percent in 2023, and recovered to 5.4 percent in 2025 and 4.6 percent year-to-date in 2026. Corporate bond issuance averaged \$1,685 billion per year in 2015–2022 and \$1,900 billion in 2023–2025 (Observed, A), with 2026 running 29.8 percent above the prior year through August. NFC net borrowing averaged \$628 billion in 2015–2022 versus \$544 billion in 2023–2025 (Observed, A): the one aggregate that ran below its baseline, though it rebounded to \$711 billion in 2025.

Lending standards tightened sharply and then eased, as Figure 4 shows. SLOOS net tightening for C&I loans to large firms averaged 1.5 percent in 2015–2022 versus 13.1 percent in 2023–2026, easing to 4.5 percent by 2026-Q3 (Observed, A). C&I loans contracted in 2023–2025 (December year-on-year: -1.3, +0.3, -2.2 percent) before rebounding 8.6 percent year-to-date through August 2026. The tightening coincides with the rate-hike cycle; the survey cannot separate AI-linked supply shifts from monetary tightening.

The time-series regressions, reported in Table 2, find no significant association between the 10-year Treasury yield and any of the three aggregates over 2015–2025, and the Chow tests find no break at 2023 for C&I growth ( $F = 1.58$ ,  $p = 0.27$ ) or net borrowing ( $F = 2.03$ ,  $p = 0.20$ ). Issuance does show a Chow break ( $F = 21.22$ ,  $p = 0.001$ ), but with eleven observations and three post-break years the test has very low power. The residual check is more informative: fitting on 2015–2022, actual 2023–2025 issuance exceeded predictions by 4.0, 7.5, and 9.2 in-sample RMSE, while C&I growth fell short by about 2 RMSE each year. Post-2023 issuance ran well above what the rate path alone predicts, which coincides with the AI financing wave but also with the broad 2024–2025 rebound from a depressed 2022 base; the bivariate model cannot attribute the gap, and we do not.

**Table 2.** US annual regressions and break tests, 2015–2025

| Dependent variable         | 10y coef | SE    | p    | $R^2$ | Chow F | Chow p |
|----------------------------|----------|-------|------|-------|--------|--------|
| Bond issuance (USD bn)     | -34.9    | 102.8 | 0.74 | 0.01  | 21.22  | 0.001  |
| C&I loan growth (pp)       | -1.74    | 1.66  | 0.32 | 0.11  | 1.58   | 0.27   |
| NFC net borrowing (USD bn) | -79.5    | 61.3  | 0.23 | 0.16  | 2.03   | 0.20   |

**Notes:** OLS of each annual aggregate on the 10-year Treasury yield,  $n = 11$ . GDP growth and default-rate controls are unavailable in the archived sources. Chow tests  $F(2,7)$  for a break at 2023; with three post-break years power is very low. Coefficients are descriptive, not structural. **Source:** Authors’ computations from Federal Reserve and SIFMA series.

### 6.3 Euro-area: quantities slowed modestly, prices did not rise

Euro-area MFI loans to non-financial corporations grew at a mean year-on-year rate of 1.99 percent in 2015–2022 and 1.67 percent in 2023–August 2026 (Reconstructed, B), with the stock rising from €5,116 billion at end-2022 to €5,463 billion in August 2026 (Observed, A). The sector detail in Figure 5 shows the divergence. Real estate plus professional and administrative services, 41.7 percent of loan stocks, kept growing at 3.3 percent after 3.9 percent; energy, water, and waste held near its earlier pace. Manufacturing loan stocks contracted from €700 billion at end-2022 to €656 billion at end-2025 (Observed, A), with mean growth of -1.7 percent after +1.7 percent; wholesale and retail trade also turned negative. Construction, after a long pre-2023 contraction (-4.2 percent mean), flattened at +0.2 percent.

The panel regression in Table 3 puts a number on the aggregate shift: the post-2023 coefficient is -3.68 percentage points (SE 3.98,  $p = 0.36$ ), negative but not statistically distinguishable from zero, with the eight-cluster caveat making inference fragile regardless. The interactions show the sectoral divergence: manufacturing times post-2023 is -1.85 (SE 0.56,  $p = 0.001$ ) and construction times post-2023 is +5.94 (SE 0.56,  $p < 0.001$ ), each measured against the pooled six-sector reference group. These are divergence facts, not causal estimates: the 2023–2025 window is the ECB hiking cycle, and the specification cannot separate tighter money from AI-linked reallocation.

On prices, Figure 6 shows the euro-area result most at odds with the crowding-out hypothesis. The NFC new-business lending rate rose from a 1.77 percent mean in 2015–2022 to 4.32 percent in 2023–2026 (Observed, A), but the reconstructed credit premium over the AAA 5-year par yield was flat: 1.98 versus 1.86 percentage points (Reconstructed, B). Higher borrowing costs in 2023–2026 were matched by higher risk-free yields; the margin banks charged over the risk-free rate did not widen. Bank lending survey standards did tighten (1.0 to 8.5 net percent) while demand fell (+11.7 to -11.4 net percent), consistent with a monetary tightening episode rather than AI-driven rationing.

**Table 3.** Euro-area sector loan-growth panel, 2015–2025

| Term                             | Coef  | SE   | p      |
|----------------------------------|-------|------|--------|
| Post-2023                        | -3.68 | 3.98 | 0.36   |
| AAA 5y yield                     | 1.04  | 1.68 | 0.54   |
| Manufacturing $\times$ post-2023 | -1.85 | 0.56 | 0.001  |
| Construction $\times$ post-2023  | 5.94  | 0.56 | <0.001 |

**Notes:** Dependent variable is sector annual mean of quarterly year-on-year loan growth, 88 sector-year observations, eight NACE sectors, sector fixed effects, SE clustered by sector (eight clusters; inference is fragile). The rate control is the AAA 5-year par yield, a weak proxy for the policy rate. Interactions measure divergence from the pooled reference group, not causation.  $R^2 = 0.33$ . **Source:** Authors’ computations from ECB series.

## 6.4 Spreads narrowed while AI capex accelerated

The US spread reconstruction tells the same story as the euro premium. US investment-grade spreads averaged 1.34 percentage points in 2021–2022 versus 1.04 in 2023–2025; high-yield spreads averaged 4.37 versus 3.56 (Reconstructed, B; the SIFMA yield history starts in 2021-Q2, so no earlier baseline exists). Euro high-yield OAS, available only from September 2023, fell from 4.37 to 2.79 percentage points by September 2026 (Observed, A). Correlations of annual AI capex growth with annual spread changes are negative but insignificant on tiny samples (US  $n = 5$ ; euro  $r = -0.44$ ,  $p = 0.18$ ,  $n = 11$ ). The sign pattern runs against the crowding-out hypothesis: spreads narrowed as capex accelerated. With the available samples, that is a fact about comovement, not evidence about causation.

## 6.5 Where the marginal credit went

The marginal-absorption arithmetic shows a growing pie with shifting shares, not a fixed pie being divided. US corporate bond issuance annual means rose from \$1,685 billion (2015–2022) to \$1,900 billion (2023–2025); applying the available 2023-Q3–2026-Q2 sector shares (Reconstructed, C, short window) puts financials at 46.8 percent and high technology at 9.9 percent of the later window. In euro-area bank lending, the €161.9 billion rise in NFC loan stocks from 2022-Q4 to 2025-Q4 was dominated by real estate plus professional and administrative services (€214.0 billion, offset by declines elsewhere), while manufacturing stocks fell €43.8 billion (Observed, A). Shared National Credit commitments rose \$1,732.9 billion from 2021 to 2025 (Observed, A), led by financials excluding banks (+\$380.9 billion) and technology/telecom/media (+\$296.6 billion, reaching \$1,113.8 billion in 2025, the largest non-financial sector).

The United Kingdom, shown in Figure 7, illustrates the aggregate-versus-channel distinction. Net M4 bank lending to the private sector rebounded to £142.9 billion in 2025, the strongest since the 1998 start of the series (Observed, A), while net sterling bond issuance by private non-financial corporations was negative every year from 2021, averaging about minus £17.1 billion in 2023–2026 (Observed, A). The bond-market contraction predates the datacenter financing wave, so it cannot be its consequence. The Bank of England’s agents reported in September 2026 that credit supply “continues to outstrip demand” while noting banks have “less appetite to lend to firms at risk of revenue erosion from AI” (Observed, A): selective reallocation by credit quality, not quantity rationing.

## 6.6 The private-credit and off-balance-sheet channel

Figure 8 shows where the AI financing is most visible: private credit. Global private credit outstanding was almost \$2.5 trillion as of 2025 (BIS, Observed, A). US technology firms’ borrowing from private credit rose from about \$22 billion (22 percent of direct lending) in 2010 to over \$1 trillion (about 44 percent) in 2025 (BIS, Observed, A). The datacenter-specific share within that is not published; industry estimates put global datacenter securitization at \$30 billion in 2025 (S&P, Observed, B) and project over half of the \$1.5 trillion datacenter expansion to 2028 coming from private credit (Morgan Stanley, Assumed, C).

The Asian deal record shows the mechanism concretely: AirTrunk’s JPY 191.6 billion Tokyo green loan (March 2026, 12 banks) and \$2.325 billion Johor financing (July 2026, 30 lenders), Princeton Digital Group’s \$750 million sustainability-linked facility with up to \$5 billion more planned, and sovereign-wealth and pension equity from GIC, ADIA, Temasek-linked vehicles, and Aware Super (all Observed, B). Banks arrange; risk distributes to private credit funds and insurers downstream. This financing is invisible in national credit aggregates because it books at project or SPV level, which is a documented finding about measurement, not a data failure.

The creditor side shows allocation shifts at the investor level rather than the bank level. GIC committed to a four-deal string of digital-infrastructure investments from 2024 to 2026; Temasek-linked ST Telemedia agreed to sell 82 percent of ST Telemedia Global Data Centres at S\$13.8 billion enterprise value to a KKR-led consortium in February 2026; Aware Super’s digital-infrastructure holdings passed A\$6 billion (all Observed, B). No primary evidence was found of insurers shifting annual-report alternative-asset buckets into digital infrastructure.

## 6.7 Cross-currency transmission

The cross-currency channel is real but, so far, absorbed. BIS data show US non-financial corporations’ euro-denominated international bond issuance actually running lower in 2023–2025 than in 2020–2022 on both gross (\$88 billion to \$71 billion per year) and net terms (Observed, A); the hyperscaler euro surge is concentrated in 2025–2026 and is recent relative to that window. The ECB’s August 2026 assessment finds hyperscalers at just under 10 percent of gross new euro non-financial issuance with no clear crowding-out spillover yet, strong demand from pension funds and insurers (hyperscalers were 15 percent of the increase in domestic euro corporate-bond holdings in the year to March 2026), and some issuers avoiding hyperscaler pricing days (Observed, B). Transmission into the euro funding pool is therefore documented; displacement within it is not yet observed.

## 7 Limitations and robustness

Nothing in this paper is causal, and the list of what the data cannot show is long enough to state in full. First, the post-2023 indicator coincides with the fastest monetary tightening cycle in four decades, the energy-price shock unwind, and the AI capex boom; no specification separates them, so every post-2023 difference is consistent with tighter money alone. Second, the euro panel has eight clusters, which makes cluster-robust inference fragile; the sector fixed-effect block is degenerate under clustering and the joint Wald test on the interactions was dropped. Third, the US regressions have eleven annual observations and a single regressor: GDP growth and Baa default-rate controls were requested but are unavailable in the archived sources (the Moody’s H.15 histories end in

September 2016). Fourth, the euro rate control is the AAA 5-year par yield, not the ECB deposit facility rate, and the reconstructed credit premium mismatches maturity and composition between the loan composite and the par yield. Fifth, no public series isolates AI or datacenter borrowing inside sector aggregates: BSI NACE detail cannot identify the use of loan funds, and the finest ICT aggregate combines transport with information and communication. Sixth, several baselines are truncated: euro high-yield OAS starts in September 2023, SIFMA yield-to-maturity in 2021-Q2, SIFMA sector shares in 2023-Q3, and the ECB securities-issues mirror ends in March 2022, removing any 2023–2026 euro-area bond-financing comparison. Seventh, private-credit evidence is thin: the BIS tech-share series has three observations and the global stock two. Eighth, 2026 observations are partial-year or guidance and are marked non-comparable throughout.

Robustness was checked where the data allow. The euro panel result is unchanged in sign when the rate control is dropped; the manufacturing and construction interactions are identified from single clusters and should be read as divergence facts with the stated fragility. The US issuance Chow break survives exclusion of 2020 but the residual pattern does not change interpretation. The spread results use two independent reconstructions (US SIFMA-based, euro ECB-based) and agree in sign.

## Falsifiability

This is a retrospective analysis; there was no pre-registered plan. The paper’s main conclusion, that the data show sectoral divergence and channel reallocation rather than aggregate crowding out, would be proved wrong by any of the following: a sustained widening of non-tech investment-grade spreads coincident with hyperscaler supply days; a contraction in total US or euro-area corporate credit aggregates while AI financing continues to grow; bank lending survey evidence attributing tightening specifically to AI-linked balance-sheet pressure rather than rates; or secondary-market repricing of AI project debt spreading to public hyperscaler bonds. The “no crowding out yet” reading of the euro cross-currency channel would break if non-tech euro issuers begin paying a measurable new-issue premium on hyperscaler supply days, which the ECB reports has not occurred.

## 8 Conclusion

The AI infrastructure boom is the largest corporate investment program in the data we examine, and its financing is large, diversified, and increasingly cross-currency and off-balance-sheet. Against it, the corporate credit aggregates of the United States, the euro area, and the United Kingdom expanded rather than contracted, lending standards tightened and then eased with the monetary cycle, and the price of credit did not rise: spreads narrowed in the United States and the euro-area credit premium held flat. What changed is the composition of credit and the channels it flows through. Euro-area manufacturing loan stocks contracted while real estate kept growing; technology borrowers absorbed a rising share of growing US syndicated credit; and AI financing increasingly runs through private credit, project vehicles, and euro, sterling, and Canadian-dollar bond markets that sit partly outside the standard aggregates.

Two implications follow, stated without uplift. First, monitoring the AI credit cycle requires watching the channels the aggregates miss: project-level debt, SPV structures with sponsor guarantees disclosed only in footnotes, and the secondary pricing of AI project risk, where the Hyperion repricing to 94.4 cents is the first observation. Second, the distributional question is open and

unanswered by this paper: the borrowers most exposed to any future rationing are the smaller, bank-dependent firms that cannot substitute across markets, and the current data cannot see them.

This paper was produced by Kilambi Research’s quantitative research pipeline with AI assistance; all analysis was verified against source data. No person is claimed to have performed steps they did not.

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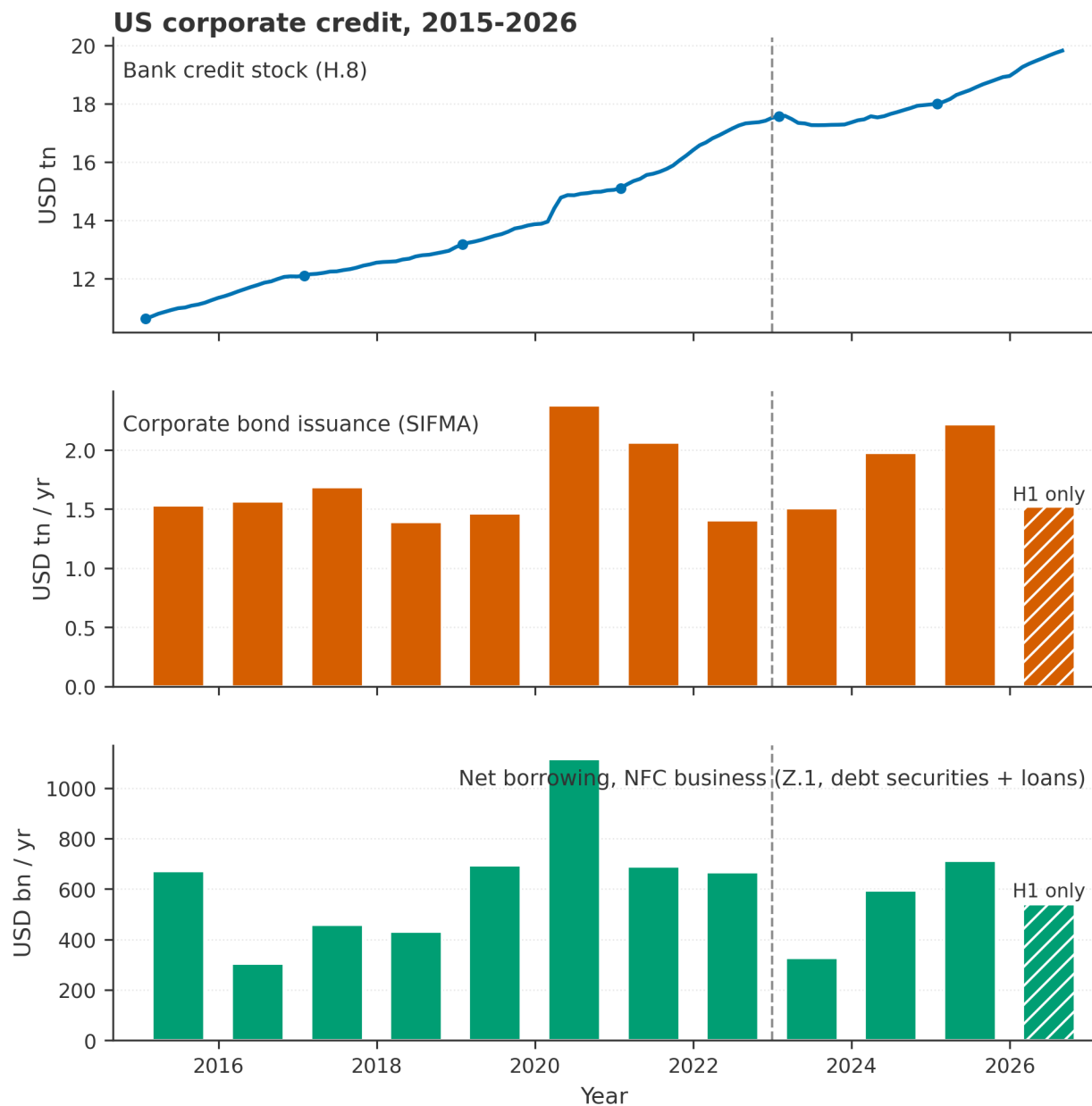
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## A Data appendix

**Table 4.** Series definitions and sources

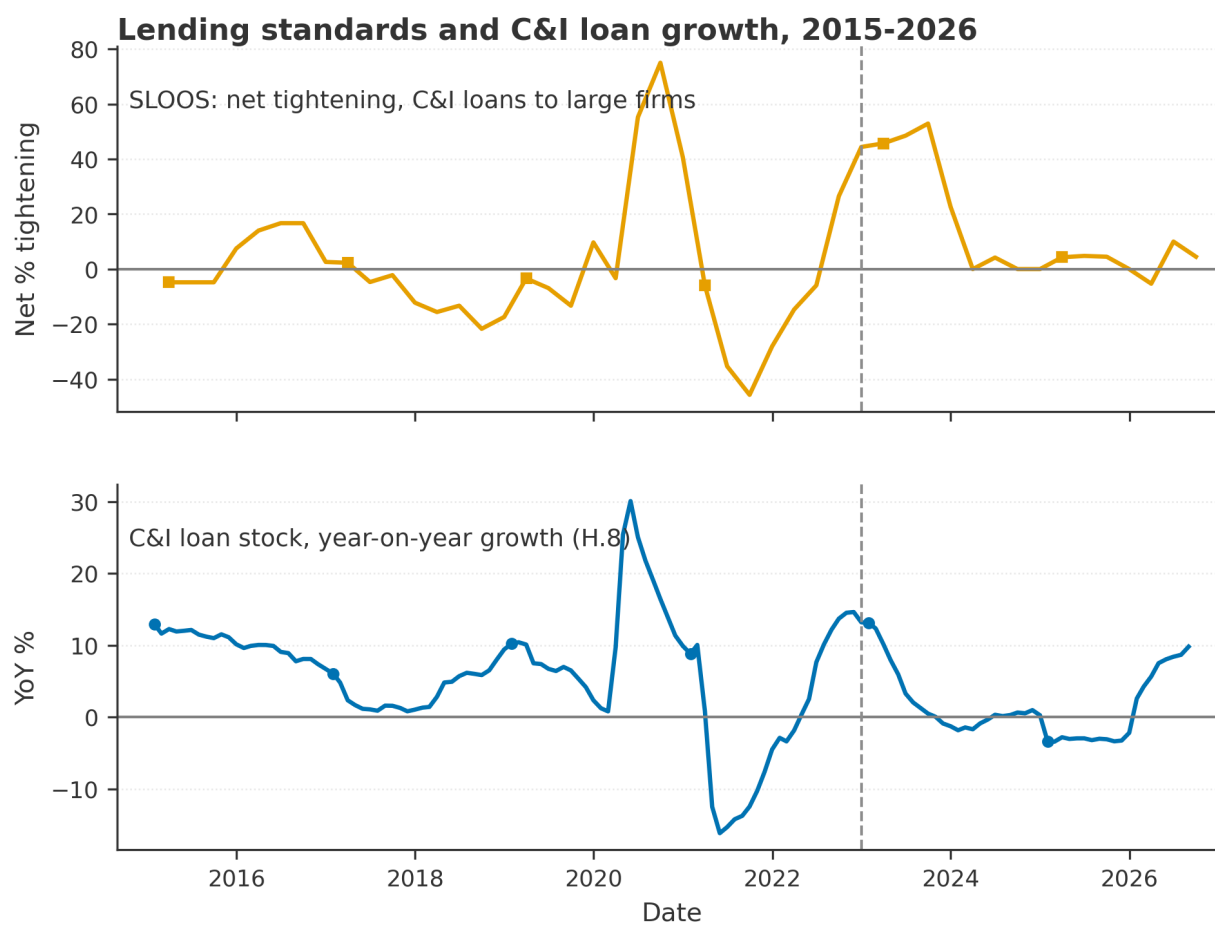
| Series                 | Source and definition                                                                          | Through                 |
|------------------------|------------------------------------------------------------------------------------------------|-------------------------|
| Hyperscaler capex      | Company 10-K/10-Q cash purchases of property and equipment; MSFT FY ends Jun 30, Oracle May 31 | 2025 (act.); 2026 guid. |
| Hyperscaler bond deals | SEC final term sheets (FWP), parsed to tranche level                                           | Sep 2026                |
| US bank credit         | Fed H.8, commercial bank credit, all commercial banks                                          | Aug 2026                |
| US C&I loans           | Fed H.8, commercial and industrial loans                                                       | Aug 2026                |
| SLOOS                  | Fed SLOOS, net % tightening, C&I loans to large firms                                          | 2026-Q3                 |
| US bond issuance       | SIFMA Research quarterly fixed-income issuance                                                 | 2026-H1                 |
| NFC net borrowing      | Fed Z.1 FL104190005, debt securities plus loans                                                | 2026-Q2                 |
| SNC commitments        | Shared National Credit program, by industry                                                    | 2025                    |
| US IG/HY spreads       | SIFMA YTM minus 10y Treasury (reconstructed)                                                   | 2026-Q2                 |
| Euro NFC loans         | ECB BSI M.U2.N.A.A20.A.1.U2.2240.Z01.E                                                         | Aug 2026                |
| Euro NACE sectors      | ECB BSI quarterly NACE estimates                                                               | 2025-Q4                 |
| Euro BLS               | ECB Bank Lending Survey, weighted net %                                                        | 2026-Q3                 |
| Euro lending rates     | ECB MIR, NFC new-business composite                                                            | Jul 2026                |
| Euro AAA 5y            | ECB AAA par yield curve, 5y                                                                    | Sep 2026                |
| Euro HY OAS            | ICE BofA Euro HY OAS via FRED                                                                  | Sep 2026                |
| Euro debt securities   | ECB SEC, NFC debt securities outstanding                                                       | Mar 2022                |
| Euro total credit      | BIS total credit to NFCs, % of GDP                                                             | 2024-Q4                 |
| US NFC euro IDS        | BIS international debt securities, US NFCs, euro de-nom.                                       | 2025-Q1                 |
| UK M4 lending          | BoE RPQB58Q, net sterling M4 lending ex intermed. OFCs                                         | Jun 2026                |
| UK PNFC issuance       | BoE CPQB3EN, net sterling securities issues by PN-FCs                                          | Jun 2026                |
| Private credit         | BIS Quarterly Review Sep 2026; IMF GFSR Apr 2024                                               | 2025                    |

**Notes:** Full series IDs, vintages, and archive paths are recorded in the workstream notes under **notes/**. Reconstructed series are flagged in the text.



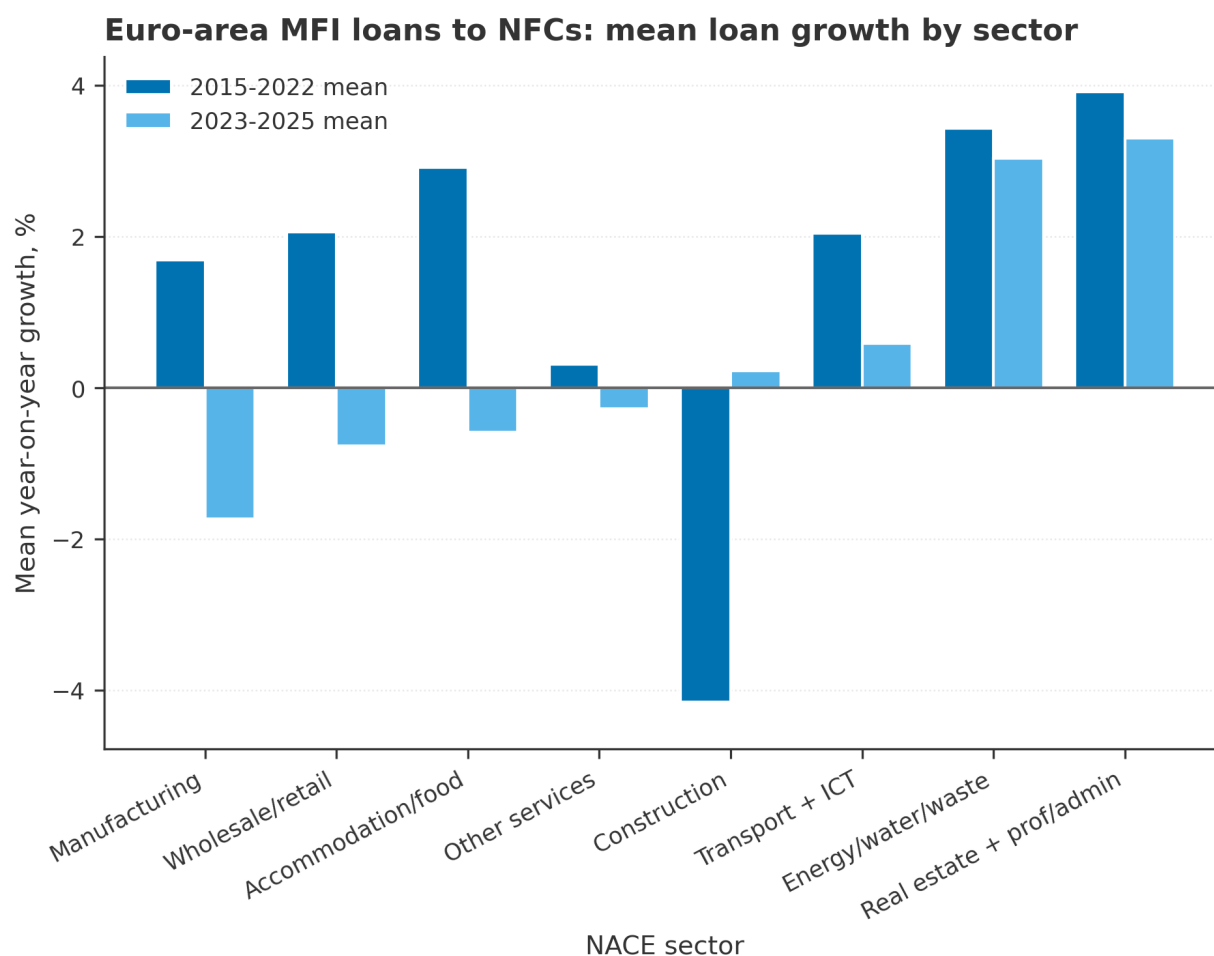
**Figure 3.** US corporate credit, 2015–2026

**Note:** Top: monthly H.8 bank-credit stock through August 2026. Middle: annual SIFMA corporate bond issuance; 2026 (hatched) is H1 only. Bottom: annual Z.1 net borrowing by nonfinancial corporate business; 2026 (hatched) is H1 only. Dashed line marks January 2023. **Source:** Federal Reserve H.8, SIFMA, Federal Reserve Z.1. All Observed, A.



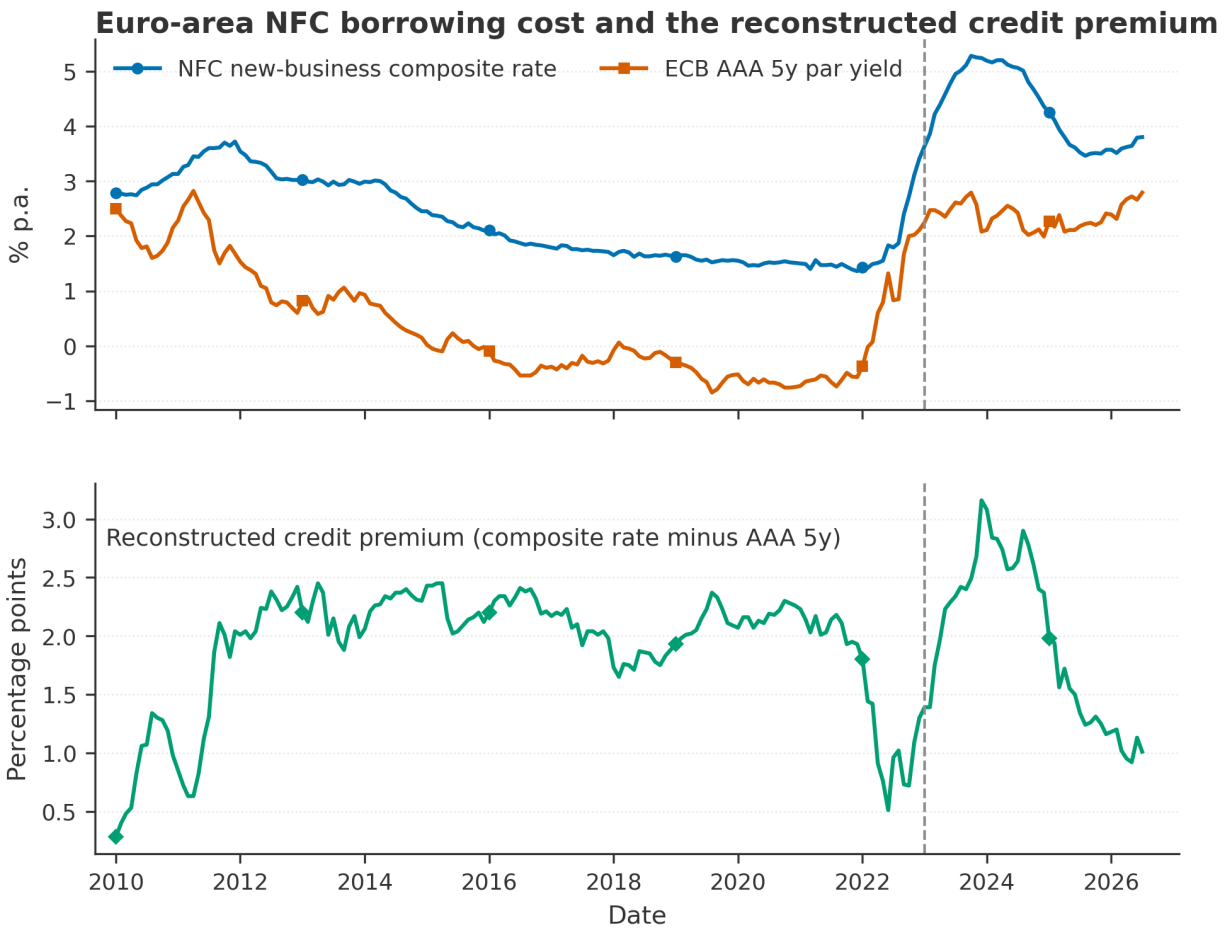
**Figure 4.** Lending standards and C&I loan growth, 2015–2026

**Note:** Top: quarterly SLOOS net tightening through 2026-Q3; positive values mean net tightening. Bottom: monthly year-on-year growth of the C&I loan stock through 2026. Zero lines shown. Dashed line marks January 2023. **Source:** Federal Reserve SLOOS and H.8. All Observed, A.



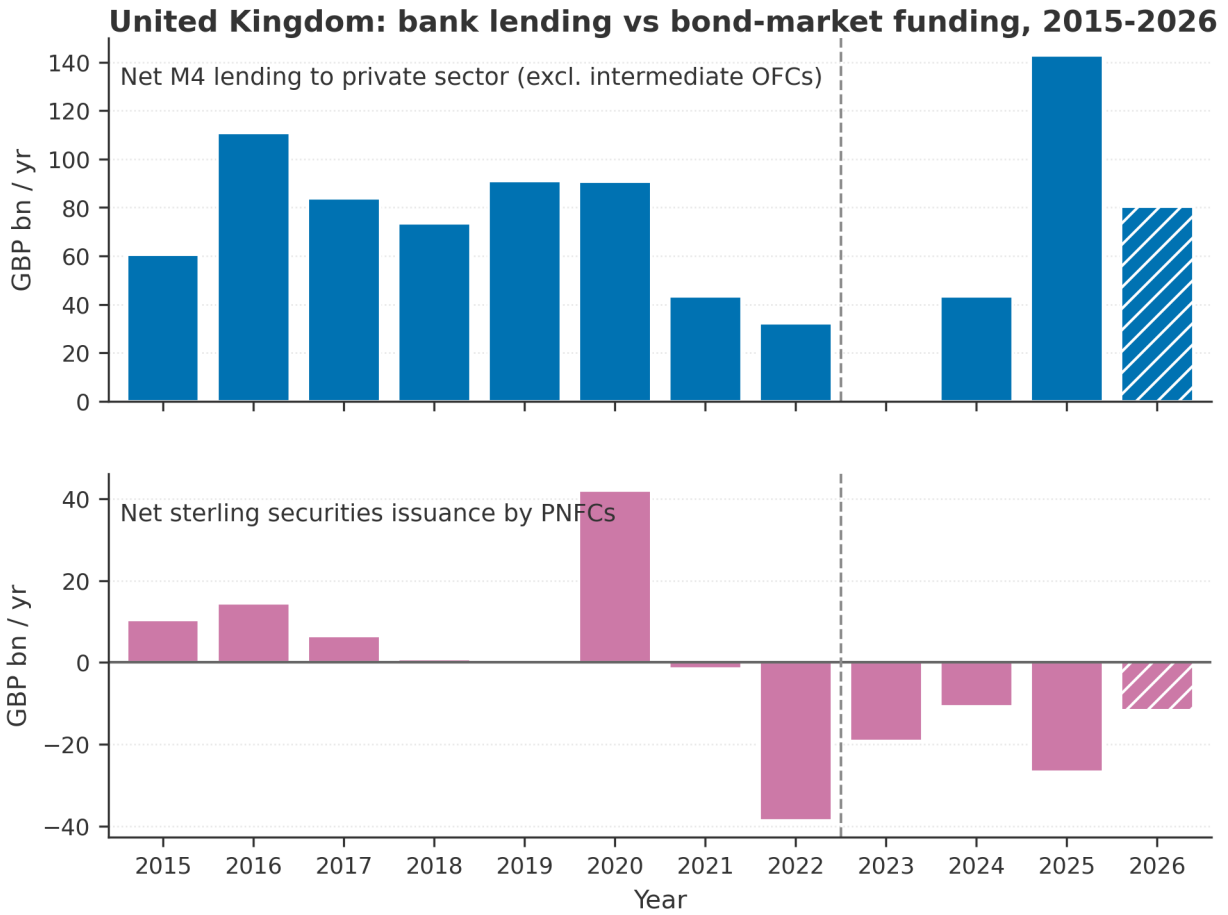
**Figure 5.** Euro-area MFI loans to NFCs, mean loan growth by sector

**Note:** Bars show the mean of quarterly unadjusted year-on-year growth rates within each window: 2015–2022 vs 2023–2025 (through 2025-Q4). Zero line shown. **Source:** ECB Statistical Data Warehouse, MFI loans to NFCs by NACE sector. Observed, A.



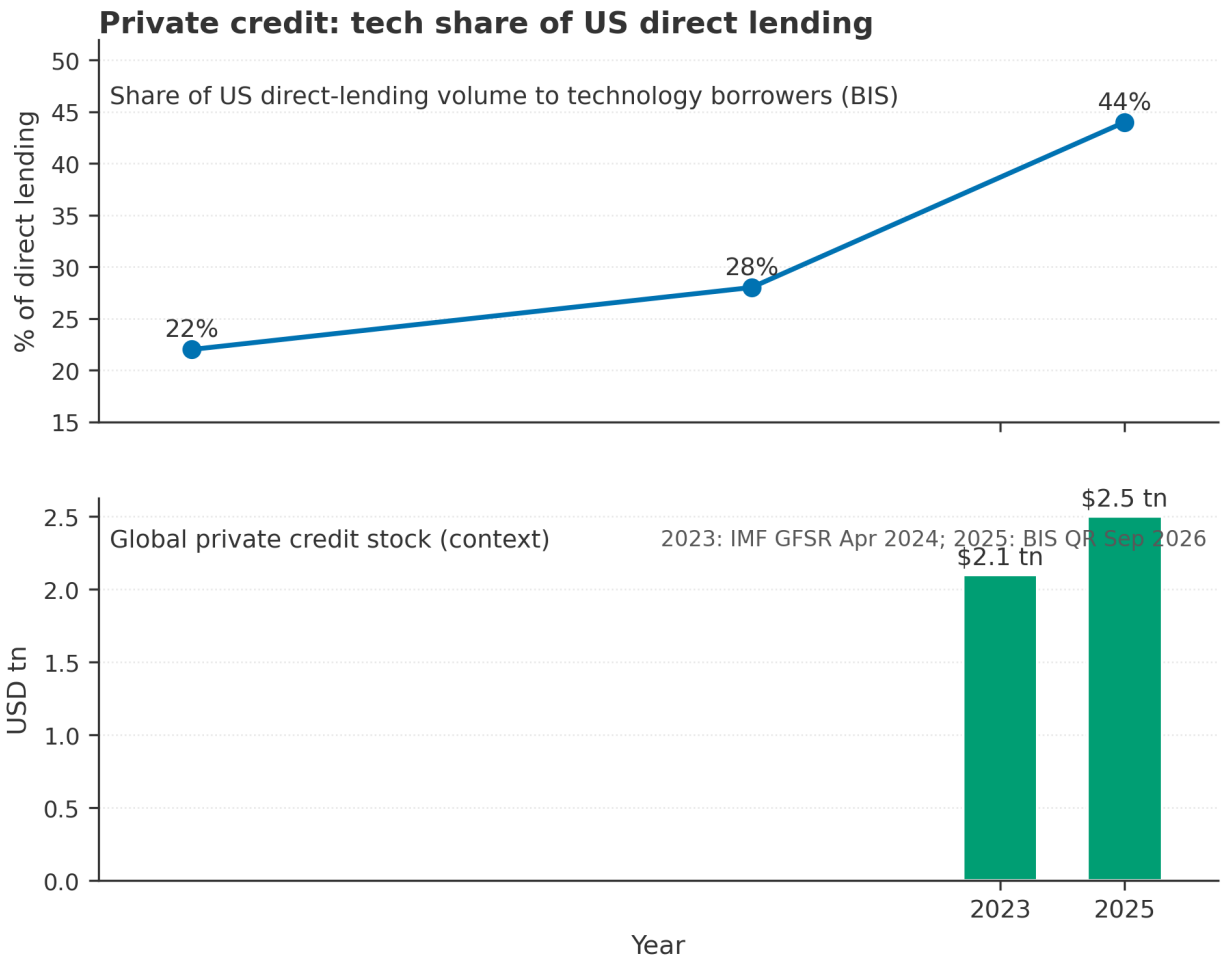
**Figure 6.** Euro-area NFC borrowing cost and the reconstructed credit premium

**Note:** Monthly, 2004–2026 (premium unavailable before 2004; the left gap in the bottom panel is missing data, not zero). Top: the two rate series; bottom: their difference. The premium is a rough proxy: the composite loan rate and the 5-year par yield differ in maturity and composition. Data through July 2026. Dashed line marks January 2023.  
**Source:** ECB Statistical Data Warehouse. Rates Observed, A; premium Reconstructed, B.



**Figure 7.** United Kingdom, bank lending vs bond-market funding, 2015–2026

**Note:** Annual net flows. 2026 (hatched) is partial-year. The near-zero 2023 M4 bar (£0.8 billion) is real, not missing. Zero lines shown. Dashed line marks the 2022/2023 boundary. M4 lending covers the private sector in aggregate, not PNFCs only. **Source:** Bank of England. All Observed, A.



**Figure 8.** Private credit, tech share of US direct lending

**Note:** Top panel has only three benchmark observations; straight segments are visual guides, not annual interpolation. The two panels use different sources and years and are shown together for context only. **Source:** BIS Quarterly Review September 2026; IMF GFSR April 2024. Observed, A.